MIL-C-17/54B
18 July 1985
SUPERSEDING
MIL-C-17/54A
15 March 1977

MILITARY SPECIFICATION SHEET

# CABLES, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 50 OHMS, M17/54-RG122

| THIS CABLE USES PVC MATERIAL AND IS NOT TO BE USED IN AEROSPACE APPLICATIONS.

NOTE: THE AIR FORCE HAS RESTRICTED THE USE OF PVC IN AEROSPACE AND GROUND SUPPORT APPLICATIONS. CABLES WITH PVC JACKETING SHALL BE USED FOR RETROFIT PURPOSES ONLY UNTIL AN ALTERNATE JACKET IS APPROVED.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-17.

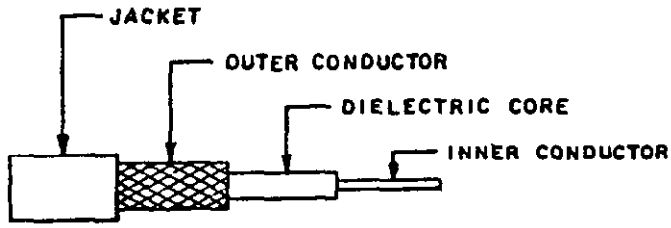



FIGURE 1. Configuration.



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TABLE I. Description.

| Component | Construction details |
|---|---|
| Inner conductor | Twenty-seven strands of tinned copper wire, each strand 0.005 inch diameter.<br>Overall diameter: 0.0308 inch ±0.0020. |
| Dielectric core | Type A-1: Solid polyethylene<br>Diameter: 0.096 inch ±0.003. |
| Outer conductor | Single braid of AWG No. 36 tinned copper wire.<br>Diameter: 0.126 inch maximum.<br>Coverage : 95.3% nominal — Alternate: 99.3% nominal<br>Carriers : 16 — Alternate: 24<br>Ends : 6 — Alternate: 5<br>Picks/inch: 12.9 ±10% — Alternate: 12.2 ±10% |
| Jacket | Type IIa: PVC.<br>Diameter: 0.160 inch ±0.005. |

ENGINEERING INFORMATION:

Continuous working voltage: 1,400 V rms, maximum.

Operating frequency: 1 GHz, maximum.

Velocity of propagation: 65.9 percent, nominal.

Power rating: See figure 2.

Operating temperature range: -40°C to +85°C.

Inner conductor properties:

(B) DC resistance (maximum at 20°C): 1.79 ohms per 100 feet.

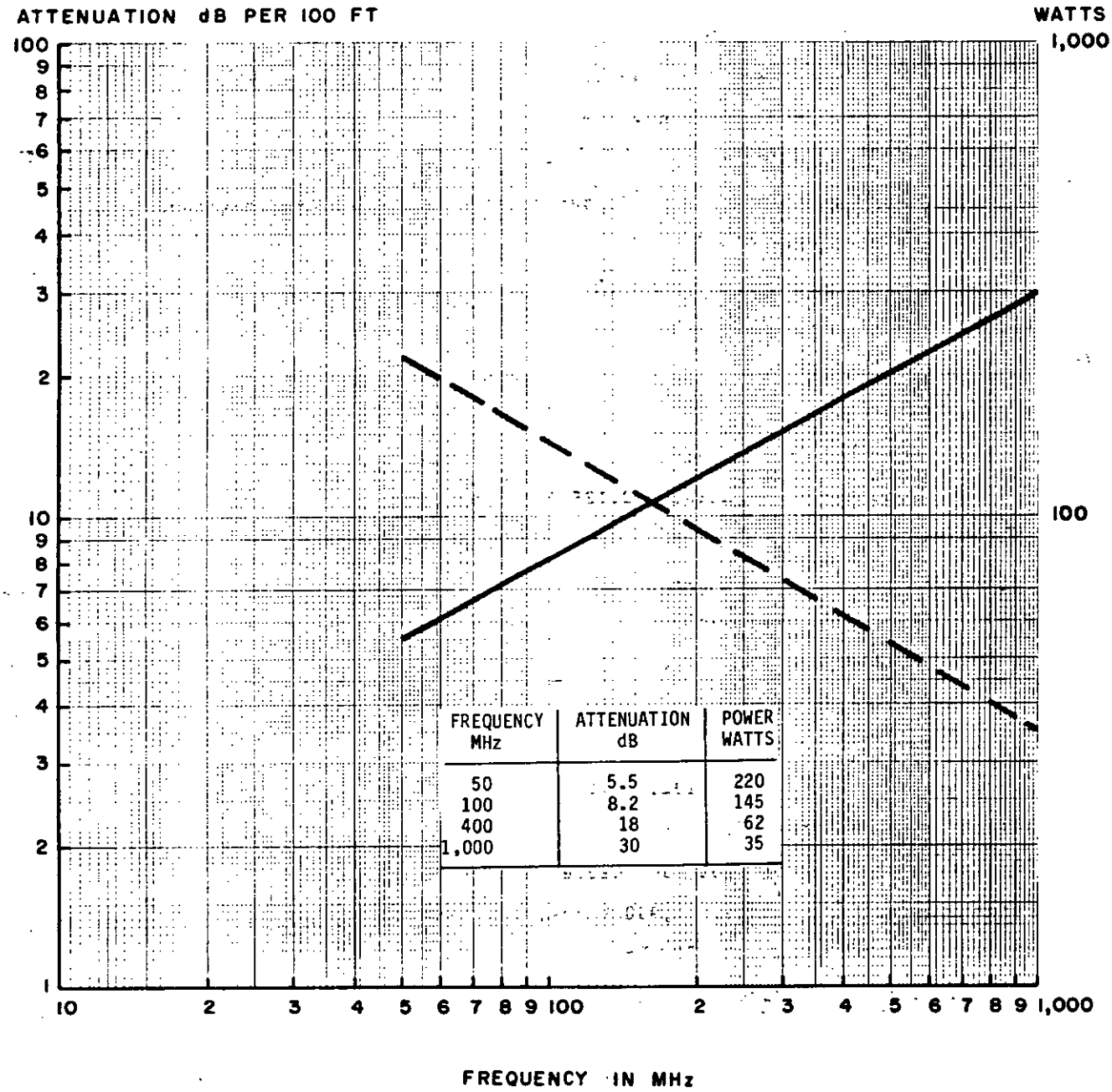
Elongation: 10 percent, minimum.

Tensile strength: Not applicable.

Engineering note: This cable is useful in general purpose medium low temperature applications. (See connector series "TNC", "BNC", and "SMA" per MIL-C-39012.)



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| FREQUENCY MHz | ATTENUATION dB | POWER WATTS |
|---|---|---|
| 50 | 5.5 | 220 |
| 100 | 8.2 | 145 |
| 400 | 18 | 62 |
| 1,000 | 30 | 35 |

FIGURE 2. Power rating and attenuation.



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REQUIREMENTS:

Dimensions, configuration, and description: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination:

Out-of-roundness: Not applicable.

Eccentricity: 10 percent, maximum.

Adhesion of conductors:

(B) Inner conductor to core: 1.3 pounds, minimum; 13 pounds, maximum.

Aging stability: +98°C ±2°C.

Stress crack resistance: Not applicable.

Outer conductor integrity: Not applicable.

(B) Cold bend: -55°C ±2°C.

Dimensional stability: +85°C ±2°C.

Inner conductor from core: 0.062 inch, maximum.

Inner conductor from jacket: 0.125 inch, maximum.

Contamination: Applicable.

Bendability: Not applicable.

Flammability: Not applicable.

Weight: 0.021 pound per foot, maximum.

Electrical:

Continuity: Applicable.

(B) Spark test: 3,000 V rms, +10%, -0%.

(B) Voltage withstanding: 5,000 V rms, +10%, -0%.

Insulation resistance: Not applicable.

Corona extinction voltage: 1,900 V rms, minimum.

Characteristic impedance: 50 ohms ±2.

Attenuation: See figure 2.

Structural return loss: See figure 3.

(B) Capacitance: 32.2 pF per foot, maximum.

Capacitance stability: Not applicable.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

Mechanically induced noise voltage: Not applicable.

Time delay: Not applicable.

Part number: M17/54-RG122.



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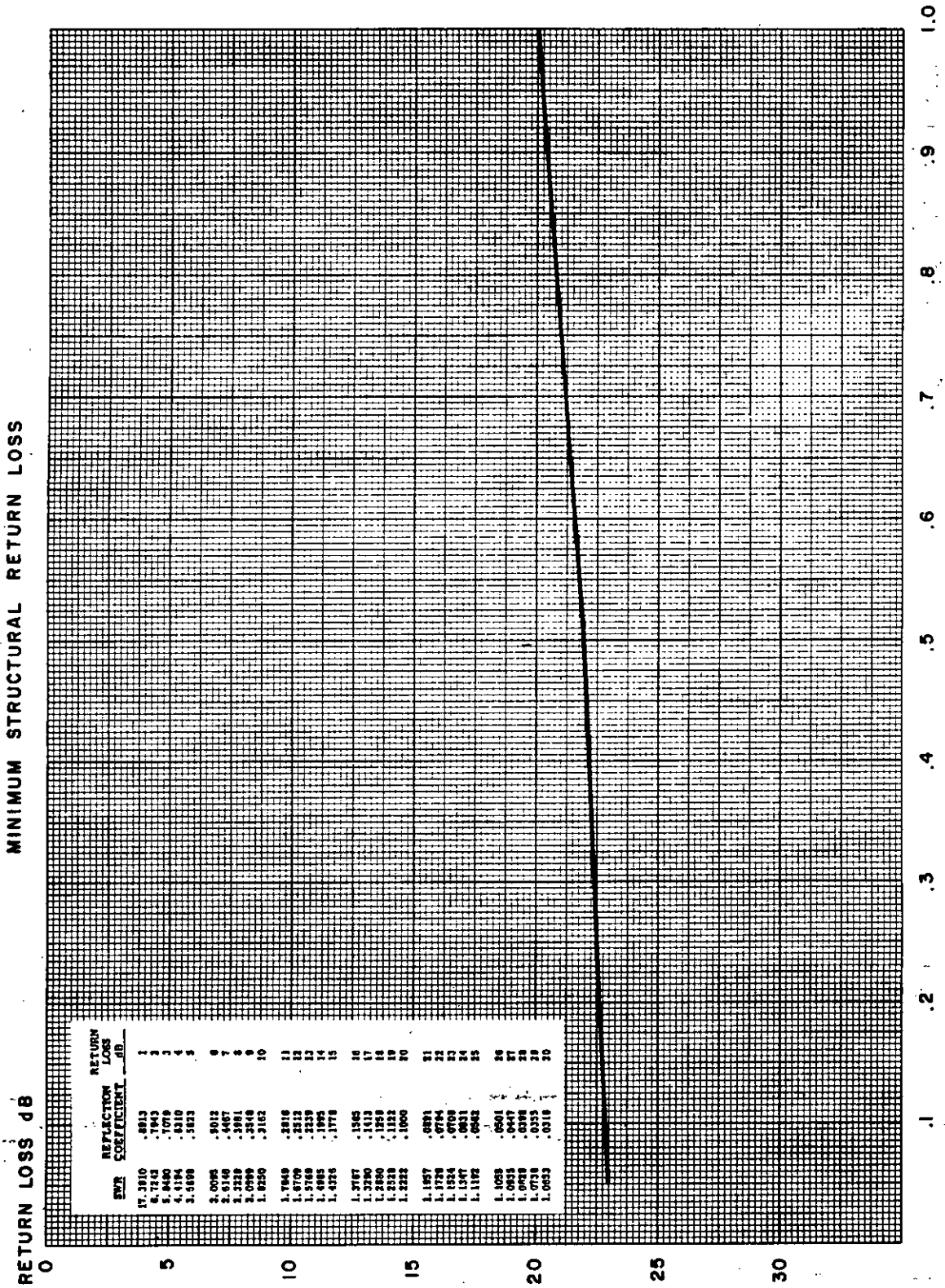



FIGURE 3. Structural return loss.

FIGURE 3.

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH, TD
Air Force - 11, 17, 99
DLA - ES, IS

User activities:
Army - AR, AT, ME
Navy - AS, MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - CR

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